

An IoT-Enabled Intelligent Classroom Monitoring Framework for Real-Time Occupancy, and Noise Detection

Abstract

Many educational institutions face challenges such as overcrowded classrooms, inefficient seating utilization and excessive noise levels, all of which can negatively impact teaching effectiveness and the learning environment. This paper presents an IoT-based Smart Classroom Occupancy Monitoring System designed to automate classroom monitoring and improve resource utilization. The system is built around an ESP32 microcontroller, which serves as the central processing unit for data collection, processing, and wireless communication. A camera module is used to count students, identify available desks, and determine classroom occupancy in real time. To enhance reliability, a radar sensor is integrated as a backup presence-detection mechanism, ensuring accurate occupancy monitoring even under poor lighting conditions or when visual obstructions affect camera performance. In addition, a noise sensor continuously monitors classroom sound levels and detects excessive noise that may disrupt the learning process. By combining occupancy monitoring and noise detection into a single platform, the proposed system provides a cost-effective and efficient solution for smart classroom management. The system aims to improve classroom discipline, optimize resource allocation, and support a more productive and technology-enabled educational environment.

Keywords

IoT; Occupancy Monitoring; ESP32; Smart Classroom; Radar Sensor; Camera Based Detection.

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